

Species-Aware Module Review: L-Lysine Catabolism through 5-Aminovalerate in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module scope:** Four-reaction Dav pathway L-lysine → 5-aminovaleramide → 5-aminovalerate → glutarate semialdehyde → glutarate **Local bucket:** KEGG ppu00310 "Lysine degradation" (32 candidate genes, 13 primary) **Date:** 2026-07-25

1. Executive Summary

The four-reaction Dav module (DavB → DavA → DavT → DavD) that converts L-lysine to glutarate via 5-aminovalerate is **fully covered and experimentally validated** in *P. putida* KT2440. All four reaction roles map to characterized, adjacent-operon genes with **direct target-organism evidence**: `davB` (PP_0383), `davA` (PP_0382), `davT` (PP_0214), `davD` (PP_0213). This is the primary/major L-lysine catabolic route in KT2440 (Revelles et al. 2005,).

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Two curation issues dominate:

1. **DavB (PP_0383) is misannotated** in local metadata as "Tryptophan 2-monooxygenase (EC 1.13.12.3)." The experimentally supported activity is **L-lysine 2-monooxygenase (EC 1.13.12.2)** (Liu 2014; Kim 2019; Revelles 2005). Recommend EC/name correction.
2. **The candidate list is a broad-map over-inclusion.** Only 4 of 32 candidates are core module genes. ~28 belong to (a) the *parallel* D-lysine/L-pipecolate/2-aminoadipate route, (b) *downstream* glutarate catabolism (outside the module boundary), or (c) generic β-oxidation / 2-oxoacid-dehydrogenase housekeeping enzymes.

No module step is a gap or "not expected." The module is satisfiable; the work needed is scoping/annotation cleanup, not gap-filling.

2. Target-Organism Pathway Definition

Included process: Aerobic oxidative deamination of L-lysine to glutarate through the "aminovalerate (Dav) pathway":

1. L-lysine + O₂ → 5-aminopentanamide (δ-aminovaleramide) + CO₂ + H₂O — flavin-dependent monooxygenase (DavB).
2. 5-aminopentanamide + H₂O → 5-aminopentanoate (5-aminovalerate) + NH₃ — amidohydrolase (DavA).
3. 5-aminovalerate + 2-oxoglutarate → glutarate semialdehyde (5-oxopentanoate) + L-glutamate — class-III aminotransferase (DavT).
4. Glutarate semialdehyde + NAD(P)⁺ + H₂O → glutarate + NAD(P)H — aldehyde dehydrogenase (DavD).

Neighboring processes to keep separate (do NOT fold into this module):

- **Parallel D-lysine / 2-aminoadipate pathway** (KT2440 runs *both* routes simultaneously): L-lysine ⇌ D-lysine (racemase `alr`) → Δ¹-piperideine-2-carboxylate → L-pipecolate (`dpkA`) → Δ¹-piperideine-6-carboxylate → 2-aminoadipate → 2-oxoadipate → glutaryl-CoA/ glutarate.
Genes: `alr`, `amaC`/PP_3590, `amaD`/PP_3596, `dpkA`/PP_3591, `amaA`/PP_5257, `amaB`/PP_5258, PP_4108, `hglS`/PP_5260 (Revelles 2005

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Revelles 2007

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Muramatsu 2005

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This is a **sibling module**, not this one.

- **Downstream glutarate → central carbon** (explicitly outside module boundary): (i) CoA-dependent glutaryl-CoA dehydrogenation (`gcdH`/PP_0158 + CoA-transferase PP_0159); (ii) CoA-independent glutarate hydroxylation via L-2-hydroxyglutarate (`glaH`/CsiD/PP_2909 + `lhg0`/PP_2910), with `ydiJ`/PP_4493 handling D-2-HG (Zhang 2018

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- **Generic overview maps / housekeeping:** β-oxidation thiolases and hydratases (`fadB`, `bktB`, `yqeF`, `paaF`, PP_0582/2215/2217/3355) and the 2-oxoglutarate/branched-chain

dehydrogenase E2/E3 components (`sucB` , `lpd` , `lpdG` , `lpdV`) are shared central-metabolism enzymes swept in by the broad KEGG map.

Alternate names / database definitions: "aminovalerate pathway," "Dav pathway," "5-aminovalerate (AMV/AVA) pathway." KEGG ppu00310 ("Lysine degradation") is a broad overview that **conflates** the Dav module, the aminoadipate module, and downstream glutarate catabolism. MetaCyc separates "L-lysine degradation I (via cadaverine)" and the aminovalerate route; the KT2440 route corresponds to the DavBADT segment.

3. Expected Step Model (Satisfiability)

#	Reaction role	Enzyme (family)	KT2440 gene	Status
1	5-aminopentanamide formation	L-lysine 2-monooxygenase (flavin MAO family)	<code>davB</code> / PP_0383	covered (direct)
2	5-aminovalerate formation	5-aminopentanamidase (C–N hydrolase superfamily)	<code>davA</code> / PP_0382	covered (direct)
3	glutarate-semialdehyde formation	5-aminovalerate:2-OG aminotransferase (class III AT)	<code>davT</code> / PP_0214	covered (direct); paralog-ambiguity note
4	glutarate formation	glutarate-semialdehyde dehydrogenase (ALDH family)	<code>davD</code> / PP_0213	covered (direct); paralog-ambiguity note

No missing steps. No step is "not expected." Genomic organization: `davAB` (PP_0382–PP_0383) and `davTD` / `davDT` (PP_0213–PP_0214) are two separate adjacent operons — the module is encoded across two loci, consistent with the "role independent of genomic organization" scope.

4. Candidate Genes and Evidence (Core Module)

DavB — PP_0383 (Q88QV1). Role: L-lysine 2-monooxygenase (step 1). **Evidence:** *Direct, target strain*. Genetic assignment (Revelles 2005); purified-enzyme biotransformation of L-lysine to 5-aminovalerate with DavA (Liu 2014,);

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repeatedly used as the KT2440 "lysine monooxygenase" part (Adkins 2013; Kim 2019). **Caveat:** Local EC 1.13.12.3 (tryptophan 2-monooxygenase) is an **over-propagated misannotation**; correct to EC 1.13.12.2 (L-lysine 2-monooxygenase). Same flavin-amine-oxidase fold explains the confusion. **Promote to full review** to fix EC/name.

DavA — PP_0382 (Q88QV2). Role: 5-aminovaleramidase / δ -aminovaleramidase (step 2). **Evidence:** *Direct, target strain*. Revelles 2005; purified with DavB (Liu 2014). EC 3.5.1.30 in metadata is consistent. High confidence; caveat minor (the "5-aminopentanamidase" name vs. common "5-aminovaleramidase" synonym).

DavT — PP_0214 (Q88RB9). Role: 5-aminovalerate:2-oxoglutarate aminotransferase (step 3). **Evidence:** *Direct, target strain*. Named DavT with EC 2.6.1.48; genetic assignment (Revelles 2005); functions in reconstituted davDT glutarate production (Adkins 2013). **Caveat:** class-III aminotransferase overlapping with GABA transaminase (GabT); GABA- pathway paralogs can perform the reaction (Rohles 2016,).

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Keep DavT as primary; add paralog-ambiguity flag.

DavD — PP_0213 (Q88RC0). Role: glutarate-semialdehyde dehydrogenase (step 4). **Evidence:** *Direct, target strain*. Genetic assignment (Revelles 2005); davDT reconstitution (Adkins 2013). EC 1.2.1.- (NAD(P)+-dependent ALDH). **Caveat:** metadata primary bucket is ppu00350, and multiple ALDH paralogs (`sad-I` /PP_2488, `gabD-II` /PP_4422, `amaB` /PP_5258, `patD` /PP_1481, `prp` /PP_2801) share broad EC space. Keep DavD as primary module assignment.

5. Gaps, Ambiguities, and Likely Over-Annotations

- **No true gaps.** All four steps covered by direct evidence.
- **DavB EC misannotation** (EC 1.13.12.3 $\rightarrow$ 1.13.12.2): highest-priority correction.

- **Bucket over-inclusion (pathway-level over-propagation):** ~28/32 candidates are not Dav-module genes. They are correctly annotated enzymes placed in the wrong *module*:
- **Sibling aminoadipate/D-lysine module:** `alr` (PP_3722), `amaD` (PP_3596), `dpkA` (PP_3591), `amaA` (PP_5257), `amaB` (PP_5258), PP_4108, `hglS` (PP_5260).
- **Downstream glutarate catabolism (out of boundary):** `gcdH` (PP_0158), PP_0159, `glaH` (PP_2909), `lhgO` (PP_2910), `ydiJ` (PP_4493).
- **Generic β -oxidation / dehydrogenase housekeeping:** `fadB`, `bktB`, `yqeF`, `paaF`, PP_0582/2215/2217/3355, `sucB`, `lpd`, `lpdG`, `lpdV`.
- **Paralog / broad-EC ambiguity** at steps 3–4: multiple class-III aminotransferases and ALDHs share EC/GO space with DavT/DavD; automated EC→gene mapping would over-call. Loss of `davTD` may be partially buffered by the GABA pathway (Rohles 2016).
- **`amaD`/PP_3596 EC ambiguity:** metadata "D-lysine oxidase (EC 1.4.3.-)" vs. Revelles 2007's functional assignment as **D-lysine dehydrogenase** — a curation-relevant discrepancy, but belongs to the sibling module.

6. Module and GO-Curation Recommendations

Module step	Recommended status
1. L-lysine 2-monooxygenase (DavB)	covered — correct EC to 1.13.12.2
2. 5-aminopentanamidase (DavA)	covered
3. 5-aminovalerate aminotransferase (DavT)	covered (paralog-ambiguity note)
4. glutarate-semialdehyde dehydrogenase (DavD)	covered (paralog-ambiguity note)

- **Module boundaries are correct for this organism** as written (module ends at glutarate; downstream CoA-dependent and CoA-independent routes excluded). No `module_needs_revision`.
- **Recommend a separate sibling module** for the D-lysine/L-pipecolate/2-aminoadipate route, which KT2440 runs simultaneously and which the candidate list currently mixes in.
- **Bucket cleanup:** demote the ~28 non-Dav candidates from this module's satisfiability set (retain as pathway-context, not module members).

- **GO terms:** ensure DavB carries GO:0050048 (L-lysine 2-monooxygenase activity), not a tryptophan-monooxygenase term; DavA GO:0047622 (5-aminovaleramidase); DavT GO:0019481-type / EC 2.6.1.48 transaminase activity; DavD glutarate-semialdehyde dehydrogenase activity. No new GO term requests appear necessary.

7. Genes to Promote to Full fetch-gene Review

1. **PP_0383 / davB** — priority: fix EC 1.13.12.3 → 1.13.12.2 and name; confirm GO.
2. **PP_0213 / davD** and **PP_0214 / davT** — confirm primary bucket (davD currently ppu00350), attach paralog-ambiguity note, verify NAD(P)⁺ cofactor preference for davD.
3. **PP_0382 / davA** — light-touch: confirm synonym normalization (5-aminovaleramidase).
4. (Sibling-module, if that module is curated) **PP_3596 / amaD** — resolve oxidase vs. dehydrogenase discrepancy (Revelles 2007).

8. Key References

- Revelles O, Espinosa-Urgel M, Fuhrer T, Sauer U, Ramos JL. **Multiple and interconnected pathways for L-lysine catabolism in *Pseudomonas putida* KT2440.** *J Bacteriol* 2005.

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— Foundational; genetic + ¹³C/¹⁵N-tracing assignment of davB/davA and both parallel routes. *Direct target-strain evidence.*

- Liu P, et al. **Enzymatic production of 5-aminovalerate from L-lysine using L-lysine monooxygenase and 5-aminovaleramide amidohydrolase.** 2014.

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— Purified KT2440 DavB + DavA. *Direct.*

- Adkins J, Jordan J, Nielsen DR. **Engineering *E. coli* for renewable production of 5-aminovalerate and glutarate.** 2013.

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— davBA + davDT reconstitution. *Parts from KT2440.*

- Kim HT, et al. **Metabolic engineering of *C. glutamicum* for glutaric acid.** 2019.

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— davB = lysine 2-monooxygenase, davA = δ -aminovaleramidase, davT/davD.

- Rohles CM, et al. **Systems metabolic engineering of *C. glutamicum* for 5-aminovalerate and glutarate.** 2016.

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— GabT/GSA-DH paralog redundancy for steps 3–4.

- Revelles O, Wittich RM, Ramos JL. **Identification of the initial steps in D-lysine catabolism in *P. putida*.** 2007.

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— Sibling D-lysine route (amaC/amaD).

- Muramatsu H, et al. **DpkA — Δ^1 -piperidine-2-carboxylate/ Δ^1 -pyrroline-2-carboxylate reductase.** 2005.

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— Sibling route (dpkA).

- Zhang M, et al. **Increased glutarate production by blocking glutaryl-CoA dehydrogenation and an L-2-hydroxyglutarate pathway.** 2018.

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— Downstream glutarate catabolism (gcdH; CsiD/glaH + lhgO). *Direct target-strain.*

Evidence character

- **Direct experimental (KT2440):** all four module steps (Revelles 2005; Liu 2014; Zhang 2018).
- **Homology / heterologous parts:** engineering studies transferring KT2440 genes into *E. coli*/*C. glutamicum* corroborate function (strong transfer, same genes).
- **Inferred:** paralog redundancy at steps 3–4 (strong mechanistically; from *C. glutamicum*).

Open questions for experts

- NAD⁺ vs NADP⁺ preference of DavD (PP_0213) in KT2440.
- In-vivo contribution of GABA-pathway paralogs to Dav flux under `davTD` loss.
- Correct EC/functional class of `amaD`/PP_3596 (oxidase vs. dehydrogenase) — sibling module.

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